

Reinforcement Learning from Human Feedback as Thermodynamic Entropy Minimisation: A Formal Isomorphism and Testable Predictions

Research Note — Version 1.0

Morten Magnusson

Symbiose Research, Sandnes, Norway

ORCID: [0009-0002-4860-5095](https://orcid.org/0009-0002-4860-5095)

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Abstract. We demonstrate that Reinforcement Learning from Human Feedback (RLHF) is formally isomorphic to statistical-mechanical entropy minimisation. The reward signal maps to negative thermodynamic energy; the policy distribution to a Boltzmann ensemble; the temperature hyperparameter to inverse temperature $\beta = 1/kT$; and the KL-divergence penalty in Proximal Policy Optimisation (PPO) to a free-energy constraint. Under this mapping, alignment corresponds to convergence toward a low-entropy, low-free-energy policy attractor, while jailbreaking corresponds to thermodynamic fluctuations over a potential barrier. We derive three testable, falsifiable predictions: (i) an optimal reward temperature T^* that scales with task entropy; (ii) grokking as a first-order phase transition with a latent heat analogue; and (iii) a universal alignment–capability trade-off governed by a free-energy bound. This isomorphism is not merely analogical: it yields novel design principles for RLHF and opens a thermodynamic branch of alignment research.

Keywords: reinforcement learning, RLHF, thermodynamics, statistical mechanics, entropy, alignment, grokking, phase transitions, free energy, Boltzmann distribution.

1 Introduction

Reinforcement Learning from Human Feedback has become the dominant post-training paradigm for large language models, enabling models such as InstructGPT [1] and its successors to follow instructions, refuse harmful requests, and exhibit behaviour aligned with human values. Despite its empirical success, the theoretical foundations of RLHF remain incompletely understood. In particular:

- Why does KL-divergence regularisation prevent collapse?
- What determines the optimal degree of reward shaping?
- Why does grokking — the delayed, abrupt emergence of generalisation — occur [3]?
- Is there a principled limit to how “aligned” a model can be without sacrificing capability?

We argue that all four questions admit natural answers under a thermodynamic interpretation. The central observation is that the RLHF objective — maximise expected reward subject to a KL-divergence constraint from a reference policy — is formally identical to minimising the Helmholtz free energy $F = \langle E \rangle - TS$ in a canonical ensemble.

This is not the first observation that machine learning has thermodynamic structure. The Hopfield network [5] is a Boltzmann machine; simulated annealing [6] is explicit thermal relaxation; variational autoencoders minimise a free-energy bound. However, to our knowledge no prior work has mapped the full RLHF pipeline — including PPO clipping, reward temperature scaling, and the grokking phenomenon — onto a unified thermodynamic framework with testable predictions.

1.1 Connection to Entropy-Flow Cosmology

The present analysis is motivated in part by the Entropy-Flow Cosmology (EFC) framework [11], in which entropy gradients drive cosmic structure formation without invoking dark matter. A central EFC principle is that gradient-driven systems exhibit universal attractor dynamics: systems flow toward configurations that minimise free entropy flux, subject to boundary constraints. RLHF exhibits structurally identical dynamics at the computational level, suggesting that thermodynamic attractor behaviour may be a domain-general principle transcending physical substrate.

1.2 Scope of This Note

This note proceeds as follows. Section 2 reviews the RLHF objective. Section 3 establishes the formal isomorphism table. Section 4 develops the thermodynamic theory of RLHF. Section 5 derives three falsifiable predictions. Section 6 discusses implications for alignment research. Section 7 states reproducibility conditions.

2 Background: The RLHF Objective

2.1 Standard Formulation

Let $\pi_\theta(a \mid s)$ denote the policy (action distribution given state), $R(s, a)$ the reward function learned from human preferences, and $\pi_{\text{ref}}(a \mid s)$ the reference (supervised fine-tuned) policy. The RLHF objective in the PPO regime is [1, 2]:

$$\mathcal{J}(\theta) = \mathbb{E}_{s \sim d^\pi, a \sim \pi_\theta} [R(s, a)] - \beta_{\text{KL}} D_{\text{KL}}(\pi_\theta \parallel \pi_{\text{ref}}) \quad (1)$$

where $\beta_{\text{KL}} > 0$ is the KL-penalty coefficient and D_{KL} is the Kullback–Leibler divergence:

$$D_{\text{KL}}(\pi_\theta \parallel \pi_{\text{ref}}) = \sum_a \pi_\theta(a \mid s) \ln \frac{\pi_\theta(a \mid s)}{\pi_{\text{ref}}(a \mid s)} \quad (2)$$

2.2 Optimal Policy Solution

The optimal policy that maximises (1) is known [2] to be:

$$\pi^*(a \mid s) = \frac{\pi_{\text{ref}}(a \mid s) \exp(R(s, a)/\beta_{\text{KL}})}{Z(s)} \quad (3)$$

where $Z(s) = \sum_a \pi_{\text{ref}}(a \mid s) \exp(R(s, a)/\beta_{\text{KL}})$ is a normalisation constant.

Equation (3) is a Boltzmann distribution with energy $-R(s, a)$ and temperature $T = \beta_{\text{KL}}$. This is the foundational identity that drives the entire isomorphism.

3 The Formal Isomorphism

3.1 Mapping Table

Table 1 presents the complete isomorphism between RLHF and equilibrium statistical mechanics.

Table 1: Formal isomorphism: RLHF $\leftrightarrow$ Statistical Mechanics. All identifications are exact, not approximate.

RLHF Concept	Statistical Mechanics Concept
Policy $\pi_\theta(a s)$	Boltzmann distribution $p(a) \propto e^{-\beta E(a)}$
Reward $R(s, a)$	Negative energy $-E(a, s)$
KL penalty coefficient β_{KL}	Temperature T (with $k_B = 1$)
Reference policy π_{ref}	Prior / vacuum state
RLHF objective $\mathcal{J}(\theta)$	Negative Helmholtz free energy $-F$
Partition function $Z(s)$	Canonical partition function Z
Policy Shannon entropy $H(\pi_\theta)$	Thermodynamic entropy S
PPO clip ratio $[1 - \epsilon, 1 + \epsilon]$	Thermostat (maintains T stability)
Value function $V(s)$	Free energy $F(s) = \langle E \rangle - TS$
Advantage $A(s, a) = R - V$	Energy fluctuation $\delta E = E - \langle E \rangle$
Grokking (delayed generalisation)	First-order phase transition
Alignment	Low-entropy policy attractor
Jailbreaking	Thermal fluctuation over barrier ΔF
Constitutional AI constraints	External field shaping energy landscape

3.2 Free Energy Derivation

We now show that the RLHF objective (1) equals $-F$ exactly. Expanding the KL divergence:

$$\begin{aligned}
\mathcal{J}(\theta) &= \mathbb{E}_{\pi_\theta}[R(s, a)] - \beta_{\text{KL}} D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}) \\
&= \mathbb{E}_{\pi_\theta}[R(s, a)] - \beta_{\text{KL}} \mathbb{E}_{\pi_\theta}[\ln \pi_\theta] - \mathbb{E}_{\pi_\theta}[\ln \pi_{\text{ref}}] \\
&= \underbrace{\mathbb{E}_{\pi_\theta}[R(s, a)]}_{\text{mean reward}} + \underbrace{\beta_{\text{KL}} H(\pi_\theta)}_{\text{policy entropy}} + \underbrace{\beta_{\text{KL}} \mathbb{E}_{\pi_\theta}[\ln \pi_{\text{ref}}]}_{\text{reference cross-entropy}}
\end{aligned} \tag{4}$$

Identifying $-E \equiv R$ and $T \equiv \beta_{\text{KL}}$ we obtain $\mathcal{J} = -\langle E \rangle + TS + \text{const} = -F + \text{const}$.

Maximising $\mathcal{J}$ is thus exactly minimising F (up to a state-dependent additive constant absorbed into Z).

This is not an approximation. The equivalence is algebraically exact given the substitution $R \rightarrow -E$, $\beta_{\text{KL}} \rightarrow T$.

4 Thermodynamic Theory of RLHF

4.1 Policy Entropy and Capability

The Shannon entropy of the policy at state s is:

$$H(\pi_\theta | s) = - \sum_a \pi_\theta(a | s) \ln \pi_\theta(a | s) \quad (5)$$

High H corresponds to high temperature: the model “explores” freely and is capable of generating diverse, creative outputs. Low H corresponds to low temperature: the model is “cold”, deterministic, and tightly aligned to a narrow reward peak.

The free energy bound connects the two:

$$F = \langle E \rangle - TH \geq F_{\min} \quad (6)$$

Minimising F at fixed T yields a specific balance between low energy (high reward) and high entropy (maintained diversity). This provides a thermodynamic proof that over-regularisation (too-high β_{KL} , too-low T) necessarily reduces capability: the model freezes into a degenerate low-entropy state.

4.2 Grokking as First-Order Phase Transition

Grokking [3] is the phenomenon where a neural network achieves near-perfect training performance but near-chance test performance for many thousands of steps, before abruptly generalising.

Under the thermodynamic mapping, we interpret this as follows:

1. The model is initially in a *memorisation phase*: high-energy, low-generalisation microstate with many equivalent configurations (high degeneracy Ω , thus high entropy $S = k \ln \Omega$).
2. Continued training at low effective temperature T_{eff} slowly increases the free-energy cost of the memorisation phase relative to the generalisation phase.
3. At a critical step t^* , the free energy of the generalisation phase crosses that of the memorisation phase:

$$F_{\text{mem}}(t^*) = F_{\text{gen}}(t^*) \quad (7)$$

4. The system undergoes a first-order transition with a latent heat analogue: training loss stalls (latent period) before the generalisation attractor is reached.

The latent heat L in ordinary thermodynamics is $L = T\Delta S$. In the grokking analogue:

$$L_{\text{grok}} = T_{\text{eff}} \cdot (H(\pi_{\text{gen}}) - H(\pi_{\text{mem}})) \quad (8)$$

A larger entropy gap between phases implies a longer stall period. This is testable (see Section 5).

4.3 Jailbreaking as Thermal Fluctuation

A jailbreak is a prompt sequence p that causes an aligned model to produce outputs outside its trained attractor basin. Under the thermodynamic model:

$$\Pr(\text{jailbreak} | p) \propto \exp(-\Delta F(p)/T_{\text{eff}}) \quad (9)$$

where $\Delta F(p)$ is the free-energy barrier that prompt p must overcome. This is the Arrhenius law / Kramers escape rate for RLHF.

Equation (9) predicts:

- More “aligned” models (lower T , steeper barriers) are harder to jailbreak.
- Jailbreak probability decreases *exponentially* with alignment strength, not linearly.
- Constitutional AI [4] raises ΔF by reshaping the energy landscape, not by lowering T — a qualitatively different mechanism.

5 Falsifiable Predictions

We derive three predictions that distinguish the thermodynamic interpretation from alternatives and that can be tested empirically with existing RLHF infrastructure.

5.1 Prediction 1: Optimal Temperature Scales with Task Entropy

Claim. The optimal KL-penalty coefficient β_{KL}^* satisfies:

$$\beta_{\text{KL}}^* \propto H_{\text{task}}^{-1/2} \quad (10)$$

where $H_{\text{task}} = -\sum_a p_{\text{human}}(a) \ln p_{\text{human}}(a)$ is the entropy of the human preference distribution over outputs.

Reasoning. At optimal alignment, the policy entropy should match task entropy: $H(\pi^*) \approx H_{\text{task}}$. From the Boltzmann relation $H \sim T \ln Z$, optimal temperature scales inversely with the precision required, which scales as $H_{\text{task}}^{-1/2}$ by central limit theorem arguments on preference variance.

Test protocol. Train a fixed-capacity model with RLHF on tasks of varying measurable entropy (e.g., creative writing vs. factual Q&A vs. code generation). Plot achieved reward vs. β_{KL} for each task class. Prediction (10) implies the optimal β_{KL} will be significantly lower for creative tasks and higher for factual tasks, with the ratio tracking $\sqrt{H_{\text{factual}}/H_{\text{creative}}}$.

5.2 Prediction 2: Grokking Latent Period Scales with Entropy Gap

Claim. The number of gradient steps in the grokking stall period, Δt_{grok} , satisfies:

$$\Delta t_{\text{grok}} \propto \frac{H(\pi_{\text{mem}}) - H(\pi_{\text{gen}})}{T_{\text{eff}}} \quad (11)$$

Reasoning. By analogy with (8), the stall corresponds to a latent heat period during which the phase transition occurs. The latent heat is $L \propto T \Delta S$; the number of steps required to absorb L is inversely proportional to the per-step “heat input” (gradient signal), which scales as T_{eff} .

Test protocol. Use the modular arithmetic datasets of Power et al. [3] with varying modulus p . Estimate $H(\pi_{\text{mem}})$ and $H(\pi_{\text{gen}})$ from model output distributions before and after grokking. Test whether Δt_{grok} is linearly correlated with $(H_{\text{mem}} - H_{\text{gen}})/T_{\text{eff}}$ across task variants (correlation coefficient $r > 0.8$ would be strong evidence; $r < 0.3$ would falsify).

5.3 Prediction 3: Universal Alignment–Capability Trade-off Bound

Claim. For any RLHF-trained model, the product of alignment score A (fraction of outputs rated acceptable by humans) and capability score C (performance on held-out capability benchmark) is bounded by:

$$A \cdot C \leq 1 - e^{-(F_0 - F^*)/T} \quad (12)$$

where F_0 is the free energy of the reference model and F^* is the free energy of the globally optimal policy.

Reasoning. Both A and C are monotone functions of $-F$ but with opposing temperature dependencies. High A requires low T (sharp reward peak); high C requires high T (maintained exploration). Their product is bounded by the fraction of total free energy extracted, which cannot exceed unity.

Test protocol. Train a family of models varying only $\beta_{\text{KL}} \in [0.01, 10]$, with all other hyperparameters fixed. Plot $A \cdot C$ against β_{KL} . Prediction (12) implies a universal maximum at β_{KL}^* with monotone decline on both sides. Systematic violation of the monotone decline on the high- β_{KL} side would falsify the model.

6 Discussion

6.1 Implications for Alignment Research

The thermodynamic interpretation reframes alignment as a *phase engineering* problem rather than an optimisation problem. Current practice treats RLHF as gradient descent toward a reward maximum. The thermodynamic view reveals that this descent has inherent limits: just as a physical system cannot reduce its temperature to zero while maintaining macroscopic coherence, an AI system cannot maximise both alignment and capability simultaneously.

This suggests several design principles:

1. **Task-matched temperature.** Set β_{KL} according to Prediction 1 rather than by grid search.
2. **Annealed alignment.** Use a schedule $T(t)$ that is high during early training (capability building) and reduced during fine-tuning (alignment crystallisation) — thermodynamic annealing applied to RLHF.
3. **Landscape architecture.** Constitutional AI and RLAIIF should be understood as tools for shaping the energy landscape (raising barrier heights ΔF) rather than as mere supplementary reward signals.
4. **Jailbreak resistance is exponential.** Equation (9) implies that even modest increases in ΔF yield exponentially large reductions in jailbreak probability. This is a stronger guarantee than linear reward-shaping intuitions suggest.

6.2 Relation to Existing Frameworks

Free Energy Principle (Friston [7]). Friston’s FEP posits that biological agents minimise variational free energy. RLHF agents minimise a formally analogous Helmholtz free energy. The connection is not merely analogical: both formalisms are instances of the same maximum-entropy inference framework (Jaynes [8]).

Entropy-Flow Cosmology. EFC posits that cosmic structure forms via minimisation of entropy-gradient flux [11]. The RLHF attractor is a computational instance of the same class of dynamics, suggesting that gradient-driven free-energy minimisation is a universal substrate-independent principle.

PAC learning. Prediction (12) is reminiscent of PAC-Bayes bounds [9], but derives from thermodynamic first principles rather than combinatorial complexity arguments. The two should be reconcilable and that reconciliation is left for future work.

6.3 Limitations

1. The isomorphism is exact at the level of the optimal policy (3). The dynamics of gradient descent toward that optimum are not identical to physical relaxation dynamics and require additional modelling.
2. “Temperature” $T = \beta_{\text{KL}}$ is a scalar, whereas in physical systems temperature varies over configuration space. Extensions to spatially varying temperature (per-layer, per-head) are non-trivial.
3. The grokking prediction (11) assumes H_{mem} and H_{gen} can be measured reliably from model output distributions before full generalisation occurs. This requires careful experimental design.
4. The alignment–capability bound (12) involves F^* , the global optimum free energy, which is not directly accessible. Empirical tests must bound this quantity from below.

7 Reproducibility and Open Science

7.1 Theoretical Status

This note presents a theoretical derivation. All results follow algebraically from the RLHF objective (1) and the identification $R \rightarrow -E$, $\beta_{\text{KL}} \rightarrow T$. No numerical experiments are required to verify the isomorphism in Table 1 or the derivation in Section 3.

7.2 Empirical Predictions

The three predictions in Section 5 are designed to be testable with standard RLHF infrastructure:

- Hugging Face TRL library [10] for RLHF training
- OpenLLM leaderboard for capability benchmarking
- MT-Bench or AlpacaEval for alignment scoring
- Power et al. [3] modular arithmetic datasets for grokking experiments (publicly available)

7.3 Data and Code Availability

No datasets or code are associated with this theoretical note. Upon empirical testing of the predictions, all training scripts, model checkpoints, and evaluation code will be released under MIT license at <https://github.com/supertedai/EFC> (repository to be created upon empirical follow-up).

7.4 Preprint and DOI

This document is deposited to Figshare at:

<https://doi.org/10.6084/m9.figshare.31940535>

An arXiv preprint will follow at cs.LG / cs.AI.

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